

 B.Tech Project Presentation

LOST & FOUND PORTAL

A Simple Web Platform to Find and Return Lost Belongings on College Campus

PRESENTED BY

VINAY KUMAR YADAV

 HTML5

 CSS3

 JavaScript

 Node.js

 Express.js

 MongoDB

 JWT Token Login

 Multer Photo Upload

SPEAKER NOTES: Good morning everyone. I am Vinay Kumar Yadav. Today I am presenting my B.Tech project called 'Lost & Found Portal'. It is a simple website that helps college students find their lost items quickly using modern web technologies.

The Problem of Losing Things in College

- **Daily Problem:** Students often lose items like ID cards, notebooks, water bottles, and phones on campus.
- **Old Methods Don't Work:** Physical notice boards and WhatsApp groups are messy and hard to search.
- **Single Website Solution:** A website gives one simple place where everyone can post lost or found items.
- **Community Help:** It makes it easy and safe for students to help each other return lost items.



Going Digital

Moving from paper notices to a simple online search system where anyone can search and post lost items.

SPEAKER NOTES: In college, students often lose their items like ID cards, books, or phones. Physical notice boards and messaging groups are not organized. Our website gives one simple place for everyone to post and search for lost items.

Problem Statement

CHALLENGES

No Single Website

Lost item information is scattered across security guards, classrooms, and informal chat groups.

Slow Recovery Time

Finding an item takes days because paper notices take time to read and get noticed by others.

Hard to Search

WhatsApp groups do not let you search easily by category, date, or place where the item was lost.

Privacy & Fake Claims

Posting phone numbers on paper boards risks spam calls, and it is hard to check if a person is the real owner.

SPEAKER NOTES: We found four main problems: no single website to check, slow updates, hard to search in chat groups, and privacy risks when sharing personal phone numbers.

🎯 Project Objectives

GOALS

🗄️ One Main Database

Save all lost and found item reports in one central database system.

✍️ Two Easy Forms

Provide simple separate forms for reporting lost items and found items.

📷 Photo Uploads

Allow users to upload photos of found items to show real proof.

🔍 Quick Search & Filter

Let users quickly filter items by name, category, or college area.

🛡️ Secure User Login

Protect user accounts with secure password login using JWT tokens.

✅ Mark as Returned

Allow item owners to change the item status from 'Active' to 'Resolved'.

SPEAKER NOTES: Our goal is to build a simple website where students can report lost or found items, upload photos, search easily, and mark items as returned once they get them back.

💡 Proposed Solution

HOW IT WORKS



🛡️ Private Contact Details

Personal phone numbers are hidden from visitors and shown only to logged-in students to prevent spam.

📍 College Campus Zones

Items are categorized by simple places like Library, Computer Lab 2, Canteen, or Sports Ground.

SPEAKER NOTES: Our portal connects the finder and the owner step by step. To protect privacy, personal contact details are shown only to logged-in users.

★ Key Features

WHAT IT DOES

🔑 User Sign Up & Login

Create account easily and log in safely using secure tokens.

📄 Report Lost Item

Post details about your missing item like name, category, and place lost.

📄 Report Found Item

Post details and photo of an item you found so the owner can spot it.

📷 Photo Upload

Upload photos directly using Multer file handler tool.

🔍 Instant Search

Type keywords to search items by title, category, or location.

🔘 Post Management

Change item status to 'Resolved' or delete your post after getting item back.

SPEAKER NOTES: The key features include easy login, reporting lost or found items, uploading photos, searching by keywords, and marking items as resolved.



Frontend

HTML5: Web Page Structure
CSS3: Styling & Layout
JavaScript: User Actions



Backend

Node.js: JavaScript Server
Express.js: Web Framework



Database

MongoDB: NoSQL Storage
Mongoose: Database Helper



Security & Files

JWT: Token Login
Multer: Photo Upload
Bcrypt: Password Safety

System Architecture

3 LAYERS



FRONTEND LAYER (Browser)

HTML Forms • CSS Designs • JavaScript Fetch API sending data

HTTP Requests



BACKEND SERVER (Node.js + Express)

API Controllers • JWT Login Guard • Multer Photo Saver

Mongoose Driver



DATABASE & STORAGE (MongoDB + Server Folder)

Users Data • Items Data • Local Uploads Folder for Pictures

Data & Images

SPEAKER NOTES: The system has three main parts: the browser frontend that users see, the Express server that processes requests, and MongoDB database that saves all data.

User Module

Handles user registration, password hashing, login token creation, and profile verification.

Lost Item Module

Form handling to post lost items with date, place, item category, and description.

Found Item Module

Form handling to report found items along with Multer photo uploading.

Search Module

Search tool to filter items by category, campus place, or item title.

Item Management Module

Checks if the logged-in user created the post. Allows updating details, deleting post, or marking item as 'Resolved'.


```
// User Schema (users collection)
```

```
{
  _id: ObjectId,
  name: { String },
  email: { String, unique: true },
  password: { String, encrypted },
  phone: String
}
```

```
// Item Schema (items collection)
```

```
{
  _id: ObjectId,
  title: String,
  type: { String, enum: ['Lost', 'Found'] },
  category: String,
  location: String,
  imagePath: String,
  status: { String, default: 'Active' },
  postedBy: { ObjectId, ref: 'User' }
}
```

 **Connecting User and Item:** The postedBy field links each item post to the user who created it. This helps us show user contact info without repeating data.

Working Flow

STEPS



SPEAKER NOTES: This flowchart shows how a user uses the portal from start to finish—from logging in to posting item details, searching, contacting, and marking it resolved.



Home Page

Project Screenshot Placeholder



Login & Register Page

Project Screenshot Placeholder



Report Item Page

Project Screenshot Placeholder



Browse Items Page

Project Screenshot Placeholder



Item Details View

Project Screenshot Placeholder



User Dashboard

Project Screenshot Placeholder

Advantages

BENEFITS

Fast & Easy to Use

Simple layout so any college student can search or post items in seconds.

One Central Place

All campus lost item reports are collected in one website instead of scattered notice boards.

Saves Time

Quick search filters help students check if their lost item has been found instantly.

Photo Proof

Uploaded photos help real owners identify their lost belongings clearly.

SPEAKER NOTES: The main advantages are: it saves time, keeps all lost items in one central place, is easy to use, and uses photos for quick identification.

Limitations

CHALLENGES

Requires Internet

Users must have an active internet connection to open the website and search items.

Depends on Honest Users

The system works best when students fill correct places, real photos, and true information.

Physical Handover Needed

Initial claim is online, but the final item return must happen in person between users.

Local Folder Storage

Photos are saved in a server folder instead of online cloud storage.

SPEAKER NOTES: The limitations are: it requires internet connection, relies on honest user inputs, and final item return must happen in person.

AI Photo Matching

Automatically match lost photos with found photos using simple AI tools.

Location Map

Show loss location spots on an interactive college campus map.

SMS & WhatsApp Alerts

Send instant SMS or message alerts when a matching item is posted.

OTP Handover Verification

Send a secret OTP code to the owner's phone to confirm item return safely.

Cloud Photo Storage

Save uploaded pictures directly to online cloud storage like AWS S3.

Admin Dashboard

Give college admin staff a simple panel to view reports and website stats.



Fast Item Return

Helps students get back lost items much faster using simple online search.



One Organized Portal

Replaces messy paper notice boards with a clean and simple database.



Modern Web Tech

Built using popular and beginner-friendly web tools like Node.js and MongoDB.

The **Lost & Found Portal** provides an easy, fast, and organized way for college students to recover lost items on campus.



THANK YOU!

Questions & Feedback Are Welcome

B.TECH PROJECT PRESENTATION

VINAY KUMAR YADAV

HTML • CSS • JavaScript • Node.js • Express • MongoDB

SPEAKER NOTES: Thank you for listening to my presentation. I am ready to answer your questions now.

Q1: What is the main difference between Frontend and Backend?

Answer: Frontend is what the user sees on the screen (HTML, CSS, JavaScript). Backend is the server code that talks to the database (Node.js, Express).

Q2: Why did you choose Node.js for this project?

Answer: Node.js lets us run JavaScript on the backend server. It is fast and handles multiple user requests easily.

Q3: What is Express.js?

Answer: Express.js is a simple backend framework for Node.js. It helps us create web routes and handle HTTP requests easily.

Q4: What is a REST API?

Answer: A REST API is a way for the frontend and backend to talk to each other by sending simple JSON data over HTTP.

Q5: What are CRUD operations?

Answer: CRUD stands for Create, Read, Update, and Delete. It means adding new items, viewing items, editing items, and deleting items in the database.

Q6: Why use MongoDB instead of MySQL?

Answer: MongoDB stores data in simple JSON-like documents. It is easy to use and flexible when adding new fields.

Q7: What is Mongoose and what does `postedBy` do?

Answer: Mongoose is a helper tool for MongoDB. `postedBy` links an item post to the user account who created it.

Q8: What is JWT (JSON Web Token)?

Answer: JWT is a secure token given to a user when they log in. It proves who the user is without asking for password on every page.

Q9: What is Multer used for?

Answer: Multer is a Node.js tool used to receive and save uploaded item photos onto the server.

Q10: How do you keep user passwords safe?

Answer: We use a library called `bcrypt` to encrypt passwords before saving them in MongoDB. Passwords are never saved as plain text.

Q11: How do you stop a user from deleting someone else's post?

Answer: We check if the logged-in user ID matches the `postedBy` ID of the item before allowing delete.

Q12: What happens when an item is marked 'Resolved'?

Answer: Its status changes to 'Resolved' so it hides from main search results, but stays saved in user history.

Q13: How do you make search fast in MongoDB?

Answer: We use database indexing on fields like `category` and `location` so MongoDB doesn't have to scan every document.

Q14: Why are photos useful in a Lost & Found portal?

Answer: Photos help real owners recognize their lost items immediately and stop fake claims.

Q15: How can you deploy this website for real use?

Answer: We can host the backend on a cloud server, use cloud storage (like AWS S3) for photos, and use MongoDB Atlas for database.